HEATMAP NODE · TRACEABILITY MATRIX

HeatMap Node traceability matrix

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
Source	docs/03b-traceability.md @ b9c67bc		

REVISION HISTORY

Version	Date	Author	Change
0.5	2026-11-01	Amish Chadha	Generated by .kit/trace.py
0.4	2026-11-01	Amish Chadha	Generated by .kit/trace.py
0.3	2026-11-01	Amish Chadha	Generated by .kit/trace.py
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Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 14 user requirements, 18 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-013	PR-009	Measurement height	HMN-CAL-001, HMN-OPT-001	Mounting review with pole owners (TRL 3); side-by-side comparison of 2.0 and 2.8 m (TRL 4)	Not met**: sensors at 2.67 to 2.69 m; pedestrian-height air 0.2 to 0.8 °C warmer in strong sun (HMN-DDR-001 D2 chose 2.8 m for reach and vandal resistance; proposed, awaiting Amish)
UR-001, UR-002	PR-005	Derived heat stress	HMN-CAL-001, HMN-STM-001, HMN-OPT-001	Calculation note (TRL 3); collocation with a reference meter (TRL 4)	At risk**: input error ±0.30 °C RSS of random terms, ±0.55 °C worst case with the fan, plus a one-sided shield bias up to 0.43 °C and height bias up to 0.4 °C; calm intervals flagged; Liljegren model error unknown
UR-003, UR-004	PR-010	Fit to existing poles	HMN-CAL-001, HMN-BLD-001	Design review (TRL 3); timed trial (TRL 4)	At risk** (time): 120 degree V-saddle fits 60 to 200 mm; 37 min estimated
UR-010	PR-014	Traceable data	HMN-CAL-001	Documentation review	At risk**: the probe calibrated before fitting fits a CalRig bay but the assembled 150 mm globe does not; payload format defined
UR-007, UR-014	PR-013	Sensor head cost	HMN-CAL-001	Priced BOM, `bom/bom.csv`	Over target by USD 6.00**: USD 136.00 for the head; core with shield and adapter USD 156.00; full node USD 292.00
UR-002	PR-001	Air temperature accuracy	HMN-CAL-001, HMN-BLD-001, HMN-STM-001	Shield error estimate (TRL 3); side-by-side field comparison (TRL 4)	At risk** (restated after the independent review of HMN-STM-001 S6): 0.43 °C with the fan at 1 m/s and more, on an assumed 50 % plate-to-air coupling, a 2 s sensor lag and a 6 s fan run; the energy-balance bound is 0.79 °C, above the 0.5 °C target; 1.04 °C at 1 m/s with the fan off
UR-002	PR-002	Humidity accuracy	HMN-CAL-001, HMN-BLD-001	Datasheet; CalRig salt fixed points (TRL 4)	Met by design: SHT45 ±1.0 %RH typical
UR-001, UR-002	PR-003	Globe temperature	HMN-CAL-001, HMN-BLD-001	Calibration in CalRig (TRL 4); response estimate (TRL 3)	Met on paper: probe ±0.18 °C (±0.26 °C with a maximum-tolerance reference); 90 % response about 12 min
UR-001	PR-004	Wind speed	HMN-CAL-001, HMN-	Datasheet of the chosen anemometer	Met on paper: start-up about 0.8 m/s (typical, unverified); calm share flagged at 20 % of an interval

User requirement	Product requirement	Title	Evidence	Verification	Status
			BLD-001, BOM	(TRL 3); wind comparison (TRL 4)	
UR-006	PR-006	Reporting	HMN-CAL-001	FieldNode firmware design; airtime calculation (TRL 3)	Met on paper: 20 B payload; 23.7 s/day at SF9; 7 days in 21.5 kB; 16 min latency for 99 % of intervals
UR-005	PR-007	Power autonomy	HMN-CAL-001	FieldNode energy budget plus the sensor load in HMN-PRC-001 (TRL 3)	Met on paper: sensors and fan 35.4 mW, 35 % of FieldNode's 100 mW design allowance
UR-008	PR-008	Outdoor survival	HMN-CAL-001, HMN-STM-001	Design review; datasheets (TRL 3)	Met on paper: FieldNode interior 57.2 °C at 50 °C air with its sun shield, against 70 °C; sensor head parts within ratings (shield factor assumed); LiFePO4 charge limit to confirm (STEM S25)
UR-008, UR-012	PR-011	Wind survival	HMN-CAL-001, HMN-BLD-001, HMN-STM-001	Wind load calculation (TRL 3)	Met on paper: arm factor 9.9 on minimum yield (7.5 with the head's weight added); 0.31 mm at 20 m/s; clamp twist factor 2.5 (preload assumed)
UR-009	PR-012	Privacy	HMN-CAL-001, HMN-BLD-001	Design review of hardware and firmware	Met by design: no imaging, audio or radio scanning parts; environmental payload only
UR-012	PR-015	Mass and loading on the pole	HMN-CAL-001, HMN-BLD-001	Mass estimate (TRL 3); the D3 model's mass is in `cad/fidelity.json`	Met on paper: sensor head 2.11 kg; complete node 4.98 kg for information; largest part 290 mm
UR-005, UR-008	PR-016	Charging limit of the cell	HMN-STM-001	Datasheet of the chosen cell; FieldNode firmware (TRL 3 paper)	To verify at TRL 4: interior 57.2 °C at 50 °C air against the usual 45 °C charge limit for LiFePO4 (STEM S25); a cross-repo item for FieldNode, new in this upgrade
UR-014	PR-017	Buildable from open files	HMN-OPT-001	`python .kit/cad_check.py --export` (TRL 3)	Met by design: `cad/fidelity.json` passes at D3 for options A and B
UR-011	PR-018	Little upkeep	HMN-STM-001, HMN-OPT-001	Maintenance steps in the build plan (TRL 3 paper); field log of fouling (TRL 4)	To verify at TRL 4: fouling of the gaps and the fan by insects and dust is unknown (STEM S27); option B would need a visit every 5 to 7 days for water (STEM S31b)

Gaps

None.